2023 F=ma Exam: Problem 13

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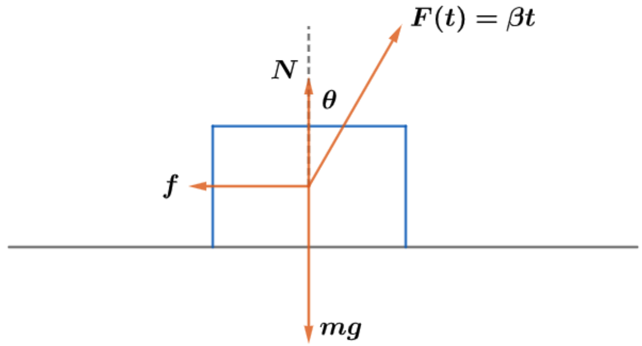

Lift off occurs when the vertical component of the external force balances the weight,

$$F(t)\cos\theta = mg$$
$$\beta t\cos\theta = mg$$
$$t_l = \frac{mg}{\beta\cos\theta}$$

Sliding occurs when the horizontal component of the external force overcomes static friction,

$$F(t)\sin\theta = \mu_s N = \mu_s(mg - F(t)\cos\theta)$$
$$\beta t\sin\theta = \mu_s mg - \mu_s\beta t\cos\theta$$
$$\beta t(\sin\theta + \mu_s\cos\theta) = \mu_s mg$$
$$t_s = \frac{\mu_s mg}{\beta(\sin\theta + \mu_s\cos\theta)} = \frac{\mu_s\cos\theta}{\sin\theta + \mu_s\cos\theta}t_l$$

Since the prefactor is less than one, $t_s < t_l$ so the block will slide first. The answer is $\boxed{\text{D}}$.